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9 December 1985
SUPERSEDING
ATPD-2111

MILITARY SPECIFICATION

HEATER TEST STAND

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a heater test stand used to test fuel-fired vehicle personnel heaters and fuel-fired engine coolant heaters.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
MILITARY

MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-52192	- Primer Coating, Epoxy.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-810D	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-45662	- Calibration System Requirements.
MS3106	- Connector, Plug, Electric, Straight, Solder Contacts, AN Type.
MS24524	- Switch, Toggle, Two Pole, Environmentally Sealed.
MS27143	- Connector, Plug, Electrical: Socket, Contact, No. 12 A.W.G. Waterproof.
MS51085	- Filter, Fluid, Pressure, Automotive Fuel (10 G.P.H. Filtration).
MS51321	- Pump, Fuel, Electrical - 24 Volts D.C. 25 G.P.H. Capacity.
AN816	- Adapter, Straight, Pipe to Tube.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

11669705	- Box Assembly, Control.
11674728	- Vehicle Receptacle Assembly.

TECHNICAL MANUAL

TM9-2540-205-24&P	- Manual for ORG, DS, GS, Repair Parts and Special Tools Lists for Vehicle Heaters.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish heater test stands which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent heater test stands delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. The heater test stand shall be constructed with 1/8-inch aluminum plate, with 1/8-inch by 1-1/2-inch aluminum or mild steel angle support (see 4.6.1 and 6.5).

3.3 Design and construction. The heater test stand shall be capable of mounting and supporting a 40-pound (lb) heater on top horizontally or on one side vertically. Military standard parts having National Stock Numbers (NSNs) shall be used whenever possible. The heater test stand shall be constructed so as to allow easy access to all components for adjustment, replacement or repair, and to facilitate setup, operation, and maintenance without the need for special tools. Welding, brazing, or soldering shall be employed only where these operations are included in fabrications of the original design. These operations shall not be employed as repair measures for defective parts (see 4.6.1 and 4.6.4).

3.3.1 Fasteners. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. Such parts shall not be swaged, peened, staked, or otherwise permanently deformed. Self-tapping screws shall not be used (see 4.6.1).

3.3.2 Configuration. The heater test stand shall conform to figure 1. A 5/8-inch diameter hole shall be located at the bottom of each corner to allow for permanent attachment to a bench (see 4.5.2. and 4.6.2).

3.3.3 Weight. The heater test stand shall weigh not more than 150 lbs (see 4.6.3).

3.3.4 Components. The heater test stand shall consist of the components specified below (see 4.6.4).

3.3.4.1 Front panel. The front panel shall be hinged and be capable of easily unlatching from the main frame and dropping to a horizontal position to allow for maintenance of indicators, pumps, wiring, and the power supply. A flexible chain at each end shall hold the front panel in place and fold inside as the front panel is returned to operating position. The front

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panel-frame interface shall be sealed against fuel seepage. All meters, switches, and lights shall be similarly sealed against the front panel. A 5-inch by 5-inch area of the front panel near the top shall be free of any components or wires, front and back, to allow for future field modifications (see 4.6.4).

3.3.4.2 Top panel. The top panel shall be hinged and be capable of being secured in a vertical position (see 4.6.4).

3.3.4.3 Tool drawer. The latchable, removable tool drawer beneath the front panel shall provide storage for removable exterior items including straps, tachometer, pyrometer probe, burette, fuel line, heater wiring harness, power cords, and drip pan (see 4.6.4).

3.3.4.4 Handles. The heater test stand shall have two handles, each capable of supporting 100 lbs. One handle shall be at each end so that the heater test stand will be two-man portable (see 4.6.4).

3.3.4.5 Curved mounts. The curved mounts shall be on the top panel, and the distance between the curved mounts shall be adjustable. The curved mounts shall be high enough to allow for placement of a drip pan (at least 1-inch high, to be provided) underneath and also to allow for exhaust connection (see 4.6.4).

3.3.4.6 Straps. The heater test stand shall have adjustable, quick-release, stainless steel straps to hold the heater in a horizontal or vertical position. The straps shall have an adjustable diameter of 6 to 9 inches and hold the heater tightly against curved mounts (see 4.6.4).

3.3.4.7 Exhaust system.

3.3.4.7.1 Flexible pipe. A 2-inch flexible pipe, at least 15 feet long, shall be provided and overpacked with the heater test stand (see 4.6.4).

3.3.4.7.2 Elbows. Three elbows shall be provided for connection to the 2-inch flexible pipe from 2-inch, 1-1/2-inch, and 1-inch heat exhaust tubes (see 4.6.4).

3.3.4.7.3 Exhaust duct. A flexible, high-temperature-resistant duct at least 48 inches long having an inside diameter of 6 inches and an appropriate quick-release clamp for attachment to the heater shall be provided with each heater test stand (see 4.6.4).

3.3.4.8 Fuel system.

3.3.4.8.1 Fuel system components. Two fuel pumps conforming to MS51321-2 shall operate in series to produce 14 pounds per square inch (psi). Each pump shall be individually controlled with a switch and indicator light. A fuel filter conforming to MS51085-2 shall protect the fuel system. The pumps and filter shall be enclosed in a separate, fuel-proof compartment inside the heater test stand. A polypropylene 100 to 200 milliliter (ml)-capacity burette with 1 ml-graduations shall be used to measure fuel flow.

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The burette shall be attached to the heater test stand with shock isolation mounts and shall be easily removable. A pressure gage capable of indicating 3 to 15 psi with an accuracy of ± 1 psi, mounted on the front panel, shall indicate fuel pressure. A kill switch having a red button 2 inches in diameter shall be located on the front panel and be capable of interrupting power to the pumps (see 4.6.4).

3.3.4.8.2 Tubing. All fuel lines shall be Tygon R-3603 plastic tubing or equivalent. Tubing shall be clear, flexible and unaffected by diesel fuel, above 32 degrees Fahrenheit (°F). Tubing shall have a inside diameter of at least 1/4 inch and shall withstand 20 psi. No tubing shall drape along the front of the heater test stand or interfere with the controls (see 4.6.4).

3.3.4.8.3 System. The main fuel pump shall receive fuel through a valve on the front panel either from a reservoir (not to be provided) and fuel filter, or from the burette. The main pump shall, through a second valve on the front panel, deliver fuel either to the heater through the booster pump and pressure gage or to the burette. No fitting is necessary for delivery to the burette. Fifteen feet of tubing shall be provided to reach the reservoir from the heater test stand. The reservoir end of the tubing shall be capable of attachment to the tube end of an adapter conforming to AN816-5-4. Connection to the heaters shall be by a 1/8-27 female dryseal PTF short fuel connector. The connector shall not drip fuel when disconnected from the heater (see 4.6.4).

3.3.4.9 Power supply. An internal power supply shall convert 115 volts (V) alternating current (ac)/60 Hertz (Hz) and 230 V ac/50 Hz to 24 ± 1 V direct current (dc). The power supply shall be capable of delivering 25 amperes (A) continuously, with no more than 1 percent (%) ripple. A set of red and black binding posts shall provide 5 A, 24 V dc for external use, protected by a 5-A fuse. A receptacle conforming to Army Drawing 11674728 shall be flush-mounted at the rear of the heater test stand and shall be capable of providing 25 A, 24 V dc to the heater test stand electrical system through a switch and 25-A circuit breaker, from an external source. The recessed area to the receptacle shall be 3-1/4 inches in diameter for cable connector clearance. The 25-A circuit breaker shall protect the power supply from shorts. The heater test stand with heater shall draw not more than 8 A at 115 V ac while in full operation. An appropriate detachable cord, 10 feet in length with safety ground, shall provide the heater test stand with electrical power. The power supply shall protect the heater test stand from, and be protected against, surges. Relays may be used if and where appropriate. Lamps shall indicate when the heater test stand main power switch is "ON" and when the external input is "ON" (see 4.6.4).

3.3.4.10 Control box. A heater control box conforming to Army Drawing 11669705 shall operate the heater and shall be flush mounted on the front panel. The box shall be modified by replacing the start/run switch with a switch conforming to MS24524-21. The box shall be rewired so that a front panel lamp will light when the heater overheat switch opens. A wiring harness, 3 feet in length with connectors at both ends, shall be provided

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to connect the heater test stand to a heater. Electrical interface with heaters shall be made by a connector conforming to MS3106R18-11S or equivalent. Electrical interface with the heater test stand shall be made on the front panel and shall be by a connector conforming to MS27143 (see 4.6.4).

3.3.4.11 Indicators. The indicators shall be mounted on the front panel and shall consist of the instruments, switches, and lamps specified below. The instruments shall include an ammeter, voltmeter, continuity indicator, clock, tachometer, and pyrometer. The switches shall be military standard toggle type and shall be similar to each other. The switches and lamps shall be permanently labeled on the front panel and grouped according to function. Panel meters shall be removable for calibration (see 4.6.4).

a. Ammeter. A 4-1/2-inch, 0 to 25, dc, rugged panel ammeter, Simpson model 3324 or equivalent, shall indicate the heater and fuel pump current draw.

b. Voltmeter. A 4-1/2-inch, 0 to 50, dc, rugged panel voltmeter, Simpson model 3324 or equivalent, shall indicate the voltage applied to the heater. A pair of red and black binding posts shall allow external use of the voltmeter through a front panel switch.

c. Continuity indicator. An audible continuity indicator shall be available for use through red and black binding posts. Approximately 50 ohms or less shall be considered continuous, whereas approximately 500 ohms or more shall not cause an audible signal. The audible signal shall continue for as long as continuity exists.

d. Clock. A digital timer/clock shall produce a one-second audible signal after a time interval preset by the operator. The timer/clock shall use easily replaced, common batteries. Both the timer/clock and the continuity indicator audible signal sound pressure levels shall be adjustable to at least 80 decibels.

e. Tachometer. A stationary, direct-reading, vibration-sensing tachometer, mounted on the heater test stand as close to the heater mount as possible, shall indicate the heater motor speed in revolutions per minute (rpm). The instrument shall have a range of 3500 to 7500 rpm with an accuracy of $\pm$ 100 rpm.

f. Pyrometer. A 4-1/2-inch, 0 to 500°F, rugged panel pyrometer, Simpson model 3324 or equivalent, shall be mounted on the front panel. An appropriate thermocouple, capable of either air or liquid measurement, shall be provided with enough detachable wire to extend 5 feet beyond the heater outlet when the heater is mounted in a horizontal position.

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g. Switches. The following switches shall be on the front panel:

1. Main power.
2. 24-V auxiliary power.
3. Main fuel pump.
4. Booster fuel pump.
5. Voltmeter circuit select.
6. Emergency/kill.

h. Lamps. The following indicator lamps shall be on the front panel, near the accompanying switch where appropriate:

1. Main power "ON" - green.
2. Accepting 24-V auxiliary power - amber.
3. Main fuel pump "ON" - amber.
4. Booster fuel pump "ON" - amber.
5. Overheat switch activated - red.

3.3.5 Interchangeability. All heater test stand items shall be interchangeable with like items (see 4.6.1 and 6.3).

3.4 Performance.

3.4.1 Operability. The heater test stand shall be capable of testing, in accordance with TM9-2540-205-24&P and the operator's manual (see 3.5), the following heaters (see 4.6.5.1).

a. Hupp models MF510A, B, and C.
b. Stewart-Warner models 10560C, M24B and B1, 10530A and B, and 939J24.
c. Precision Products model 4180.

3.4.2 Environmental.

3.4.2.1 Vibration. The heater test stand shall meet the requirements of 3.4.1 after being subjected to the vibration test specified in 4.6.5.2.1.

3.4.2.2 Shock. The heater test stand shall meet the requirements of 3.4.1 after being subjected to two drops from a height of 12 inches onto a concrete floor (see 4.6.5.2.2).

3.4.2.3 High temperature. The heater test stand shall meet the requirements of 3.4.1 while operating in an ambient air temperature of 100°F, and after storage in an ambient air temperature of 160°F (see 4.6.5.2.3).

3.4.2.4 Low temperature. The heater test stand shall meet the requirements of 3.4.1 while operating in an ambient air temperature of 32°F, and after storage in an ambient air temperature of minus 55°F (see 4.6.5.2.4).

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3.5 Operator's manual. An operator's manual shall be furnished with each heater test stand and shall contain operating instructions, drawings, illustrations, a parts list including NSN's, original manufacturer's part numbers, assembly and disassembly instructions, wiring diagrams, and safety precautions to ensure proper installation, operation and maintenance of the heater test stand. The manual shall include a pretest checklist for proper fuel and electrical heater connections. Five copies of the manual shall be provided to the procuring agency (see 4.6.4).

3.6 Surfaces, coatings, and paint. All surfaces of casting, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, spurs, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled. Painting and finishing of the heater test stand shall be in accordance with the manufacturer's best commercial practice provided the following minimum criteria are met or exceeded: All interior and exterior surfaces to be painted shall be cleaned to be free of all foreign matter detrimental to painting and at least one coat of primer conforming to MIL-P-52192 (or a phosphate or chromate base) and two coats of enamel conforming to MIL-C-46168 or equivalent (i.e., epoxy enamel) shall be applied. Unpainted metal and working surface exposed to atmospheric conditions shall be finished to resist corrosion. The type of finish applied shall be compatible with the base metal. The heater test stand shall be painted a low luster forest green. Paint shall be unaffected by diesel fuel (see 4.5.2, 4.6.1, and 4.6.2).

3.7 Identification and marking. Identification and marking shall be in accordance with MIL-STD-130 (see 4.5.2 and 4.6.2).

3.7.1 Nameplate. A corrosion-resistant nameplate shall be permanently attached to the heater test stand. All lettering shall be capitals and the numerals shall be arabic (except the model number shall be in the manufacturer's customary designation). Letters and numerals shall be of such a size as to be clearly legible. The nameplate shall show the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Manufacturer's model designation.
d. Manufacturer's serial number.
e. Contract number.
f. Stock number (use number specified in contract).
g. Date of manufacture.
h. U.S.

3.8 Wiring diagram plate. A wiring diagram plate, showing all wires and connections, shall be located on the inside of the front panel (see 4.6.4).

3.9 Warning plate. A 4-inch by 6-inch, black-on-yellow, warning plate or decal shall be located on the front panel, and shall read as follows:

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"WARNING

Do not touch heater during operation.

Properly vent exhaust. Have easy

access to fire extinguisher."

The word "WARNING" shall occupy at least 20% of the plate (see 4.6.4).

3.10 Workmanship. The heater test stand and accessories shall have a good general appearance as evidence of high caliber workmanship and shall conform to a quality of construction commensurate with the requirements specified herein and with the best quality prevailing among manufacturers normally producing equipment of the type specified herein (see 4.5.2 and 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F
b. Relative humidity 50 ± 30%
c. Barometric pressure 28.5 + 2.0 inches mercury (Hg)
   - 3.0

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4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select one heater test stand produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply heater test stands that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Test
Materials and construction	3.2, 3.3, 3.3.1, 3.3.5, and 3.6	4.6.1	X		
Defects (see 4.5.2 and table II)	3.3.2, 3.6, 3.7, and 3.10	4.6.2	X	X	
Weight	3.3.3	4.6.3	X		
Features	3.3, 3.3.4 thru 3.3.4.11, 3.5, 3.8, and 3.9	4.6.4	X		X
Operability	3.4.1	4.6.5.1	X		X
Vibration	3.4.2.1	4.6.5.2.1	X		
Shock	3.4.2.2	4.6.5.2.2	X		
High temperature	3.4.2.3	4.6.5.2.3	X		
Low temperature	3.4.2.4	4.6.5.2.4	X		

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the heater test stands of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3.2).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.2).	SIE 1/
103	Identification marking, improper (see 3.7 and 3.7.1).	Visual
104	Faulty workmanship affecting performance (see 3.10).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.2).	SIE 1/
202	Surfaces, coatings, and paint, improper preparation, application, or color (see 3.6).	Visual
203	Faulty workmanship affecting appearance (see 3.10).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Test. Each heater test stand shall be subjected to the quality conformance tests specified in table I.

4.5.4 Failure. Failure of any heater test stand to pass any of the specified quality conformance tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.6 Methods of inspection.

4.6.1 Materials and construction. Conformance to 3.2, 3.3, 3.3.1, 3.3.5, and 3.6 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3.2, 3.6, 3.7 and 3.10 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Weight. To determine conformance to 3.3.3, weigh the heater test stand.

4.6.4 Features. Conformance to 3.3, 3.3.4 through 3.3.4.11, 3.5, 3.8, and 3.9 shall be determined by exercising the heater test stand and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

4.6.5 Performance.

4.6.5.1 Operability. To determine conformance to 3.4.1, install and test one of the specified heaters, and verify the operability and accuracy of the heater test stand components.

4.6.5.2 Environmental.

4.6.5.2.1 Vibration. To determine conformance to 3.4.2.1, subject the heater test stand to the test specified in MIL-STD-810D, method 514.3, category 1, procedure 1 (omit step 6). Vibrate the heater test stand for 2 hours in accordance with figures 514.3-7 to 514.3-9. The front panel and drawer shall not come unlatched. All items capable of being stored shall be stored for the duration of the test. After the test, examine the heater test stand for damage and test as specified in 4.6.5.1.

4.6.5.2.2 Shock. To determine conformance to 3.4.2.2, subject the heater test stand to two drops from a height of 12 inches onto a concrete floor. The heater test stand shall be suspended in an upright position with an automatic quick release mechanism before each drop, with all detachable items secured in the drawer. After the test, examine the heater test stand for damage and test as specified in 4.6.5.1.

4.6.5.2.3 High temperature. To determine conformance to 3.4.2.3, test the heater test stand as specified in MIL-STD-810D, method 501.2, as follows:

a. Procedure I. 160°F for 6 hours.
b. Procedure II. Stabilize at 100°F and test as specified in 4.6.5.1.

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4.6.5.2.4 Low temperature. To determine conformance to 3.4.2.4, test the heater test stand as specified in MIL-STD-810D, method 502.2, as follows:

a. Procedure I. Minus 55°F for 6 hours.
b. Procedure II. Stabilize at 32°F and test as specified in 4.6.5.1.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. This heater test stand is intended to facilitate the testing and repair of fuel-fired vehicle personnel heaters and fuel-fired engine coolant heaters.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable drawings.
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection is not required (see 4.4).
h. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Interchangeable item. An interchangeable item is one which (1) possesses such functional and physical characteristics as to be equivalent in performance, reliability, and maintainability, to another item of similar or identical purposes; and (2) is capable of being exchanged for the other item (a) without selection for fit or performance, and (b) without alteration of the items themselves or of adjoining items, except for adjustment.

6.4 Supersession data. This military specification supersedes USATACOM Description for Purchase ATPD-2111, Heater Test Stand, dated 22 July 1985.

6.5 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged.

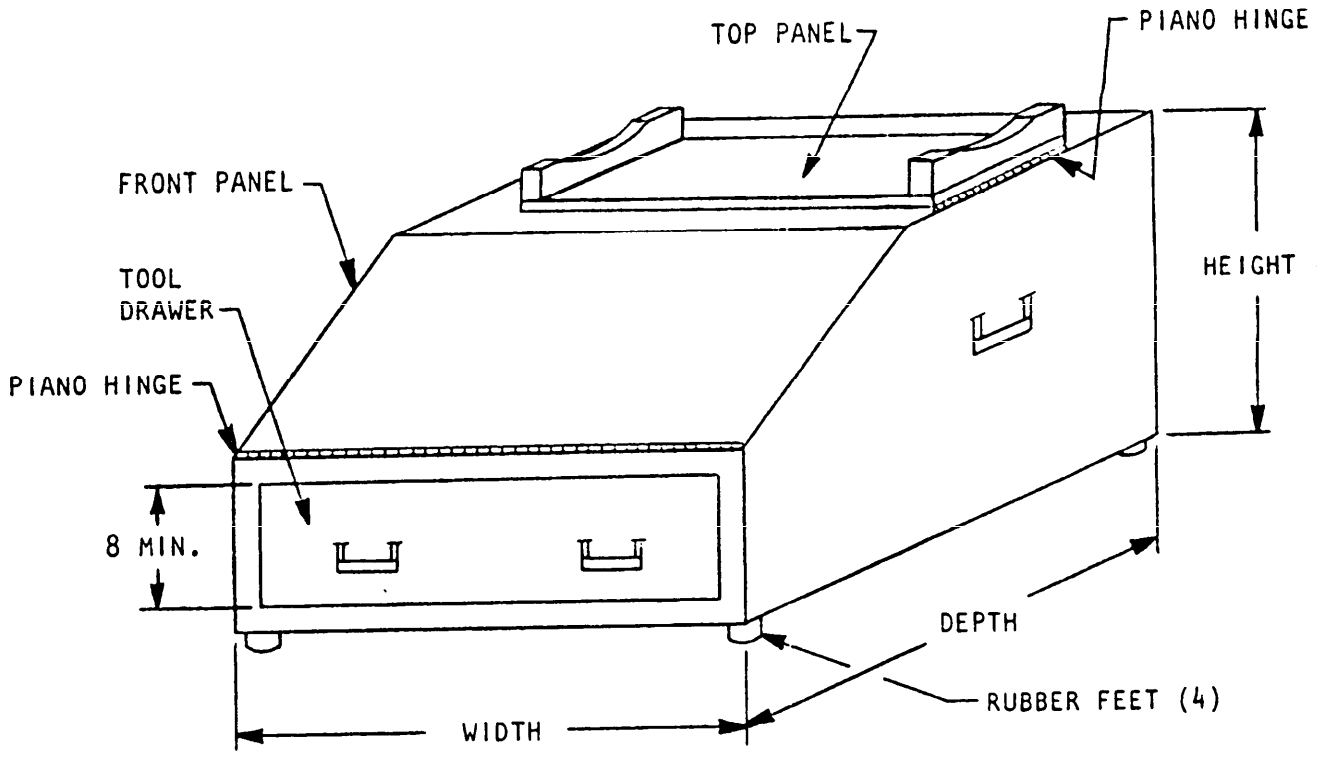
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MAXIMUM DIMENSIONS			
	HEIGHT	WIDTH	DEPTH
FRAME	28	30	24
ASSEMBLED	38	36	26



NOTE: DIMENSIONS ARE IN INCHES.

FIGURE 1. Heater test stand.

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